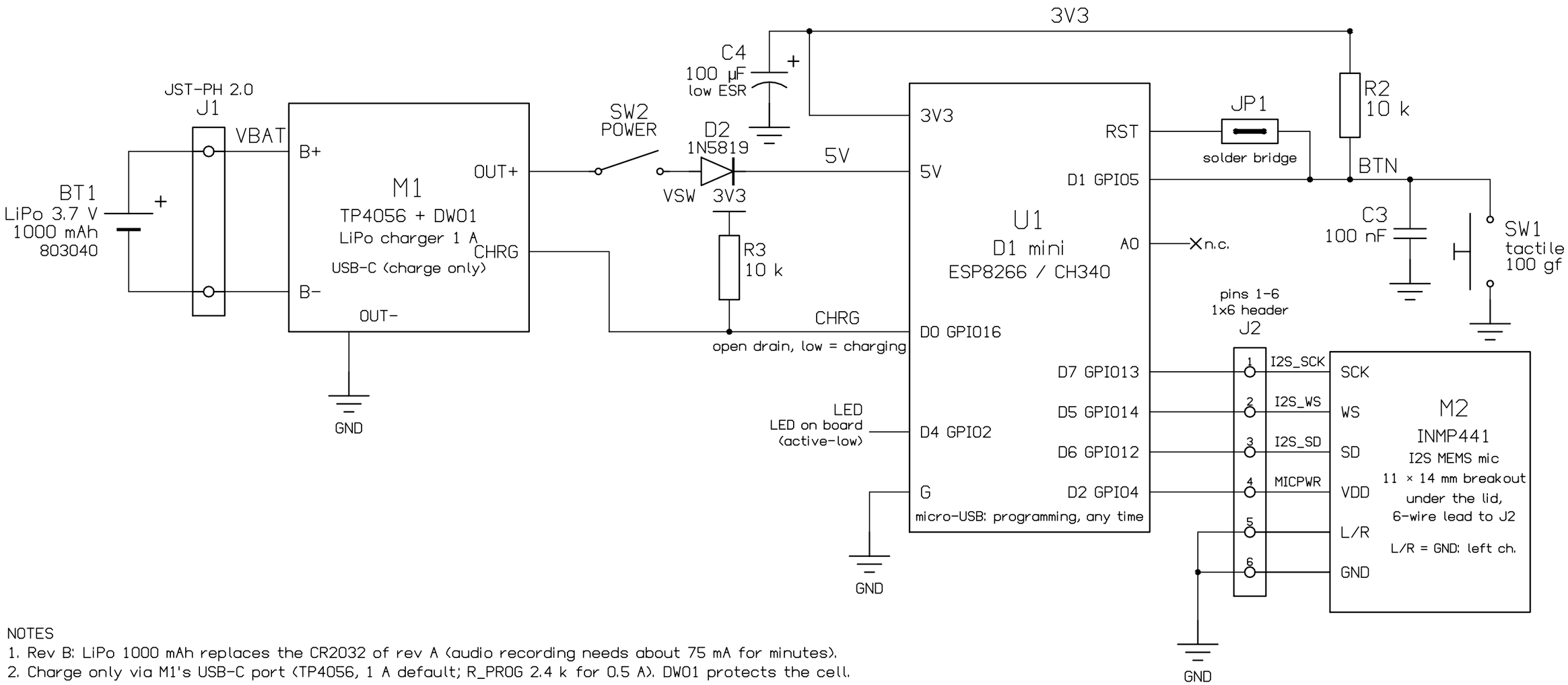


NOTES

1. Rev B: LiPo 1000 mAh replaces the CR2032 of rev A (audio recording needs about 75 mA for minutes).
2. Charge only via M1's USB-C port (TP4056, 1 A default; R_PROG 2.4 k for 0.5 A). DW01 protects the cell.
3. D2 blocks USB back-feed into the LiPo: USB programming is allowed at any time, with SW2 in any position.
4. A0 n.c.; I2S mic on GPIO12/13/14 (the ESP8266's fixed I2S RX pins): SCK = D7, WS = D5, SD = D6. J2 pins 1-6 = SCK, WS, SD, VDD, L/R, GND; L/R tied to GND (left channel).
5. M2 VDD power-gated by D2/GPIO4 (1.4 mA). M1 CHRG (open drain, low = charging) -> D0/GPIO16, R3 10 k pull-up to 3V3.
6. JP1 (solder bridge) closed = battery mode: a press pulls RST low and wakes U1 from deep sleep. Open = always_on (USB).
7. GPIO2 drives the LED already on the D1 mini (active-low): on while recording. SW1 pressed = GPIO5 LOW.
8. M2 INMP441 (11 x 14 mm) sits in a pocket under the lid, sound port over a 2 mm hole; 6-wire lead to header J2.
9. Rev C: D2 is THT (1N5819 DO-41); M1 glued with foam tape and wired from its pads; every lead/wire position is in hardware/layout/layout.json (perfboard-layout + perfboard-solder-side sheets, BUILD.md).

ESP8266 D1 mini MQTT battery button		
Rev C: LiPo, button, I2S mic Date: 2026-10-26	Rev C CADCraft	Sheet 1 / 1